diagram, 18
dynamic languages, 27
initializing with Nil Guards, 221
Lazy Instance Variable, 221, 237
safe levels, 149
self keyword, 34, 80, 86
top-level, 79–80, 100, 102
viewing, 17
instance_eval method
blocks, 85–88
Builder, 68
compared to class_eval, 108
Self Yield, 223
singleton classes, 127
Strings of Code vs. blocks, 145
instance_exec method, 86
instance_method method, 94
instance_method_already_implemented? method, 204
instance_variable_get method, 154
instance_variable_set method, 27, 154
interactive fiction, 228
internal DSLs, 229
interpreters
about, xxi
Pry, 49
REST Client, 141
introspection, 3, 7, 54
irb
Binding objects, 144
listing methods, 15
nested sessions, 144
viewing self, 35

J

Java
compile/runtime time, 8
domain-specific languages, 230
nested visibility, 79
static fields, 109–110
Java Virtual Machine, xxi
JRuby, xxi

K

Kernel Methods
ap method, 33
binding method, 143
block_given?, 74
callers method, 134
const_get method, 187
load method, 26, 95, 133, 150
local_variables method, 79
method lookup, 32
Method objects, 94
procs and lambdas, 89–93, 100, 149, 224, 234, 242
require method, 26, 133, 150
security, 150
Shared Scopes, 84
singleton_method method, 94
spell book, 237
tap method, 223
using keyword, 75–77
klass variable, 151

L

lab rat program example, 121, 123
lambda method, 89, 92
lambda? method, 92
lambdas, 88–89, 91–93, 102, 234
language constructs, 3, 7
languages
bending Ruby, xvii
domain-specific (DSLs), 97–103, 227–230
functional programming, 73
general-purpose (GPLs), 228–230
static, 45, 70
type checking, 45
Lazy Instance Variable, 221, 237
libraries, importing, 26
line argument, Binding objects, 144
listing
comma-separated example, 91
constants, 23
methods in irb, 15
load method
aliases, 133
Namespaces and, 26
security, 150
UnboundMethod example, 95
Loadable class, UnboundMethod example, 95
local_variables method, 79
Logger class example, 86

M

main object, 36, 80
make language, 228, 230
Markaby gem, 229
Mash class, 57, 59
math quiz, 136
Matz's Ruby Interpreter, xxi
memory_size attribute, 49
message of the day example, 26
meta classes, *see* singleton classes
metaprogramming
about, xvii
defined, 3, 7–8, 163
DSLs and, 230
flexibility, 188
programming as, 8, 163, 213
method execution, 28, 33–36
method lookup
direction, 30, 118, 121, 126
object model, 28–33, 55
review, 121
singleton classes, 118–122
method method, 94
Method objects, 94–96
method_added method, 158
method_body method, 210
@method_cache.compute_if_absent, 210
method_missing method
attribute methods, 200–206
avoiding duplication, 48, 55–71
Blank Slates, 66–69
Dynamic Proxies, 58–64
overriding, 56–64
quiz, 64–66
resources, 70
method_removed method, 158
method_undefined method, 158
methods, *see also* Around Aliases; Class Macros; class methods; Dynamic Methods; instance methods; Kernel Methods; method lookup; Mimic Methods; Singleton Methods; wrappers
about, 17